claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-aa1f83416163a72a9.jsonl`
- SHA-256: `ead27e2f0a7142a3fcd00c740f8be17b64df729fed66646337a9e3b8e18b3adf`
- Source modified: `2026-06-06T17:31:08+00:00`
- Imported at: `2026-07-05T16:48:28+00:00`
- project: `wf_6237b110-d58`
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`

Transcript

1. user (2026-06-06T17:30:26.467Z)

Source Extractor

```
Research question: "Research alternative methods to automatically detect rally start/end
segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-
HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation
is the best approach or whether other methods are competitive or complementary. Keep it focused
("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches
over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):
- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY
inference at runtime ? privacy lever: user video never leaves the device. It's a commercial
product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-
commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We
derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We
ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision
ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label
rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features).
Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the
learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck
(its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes /
pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd
noise, umpire/score calls).

SPECIFICALLY COVER:
1. Method families for rally/point segmentation in racket sports (badminton, tennis, table
tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep
models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance"
gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-
overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM
teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference
COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their
licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher
? local-distillation the best, or are some alternatives competitive or complementary ?
especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run
fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory +
person detector + audio)."
```

Fetch and extract key claims from this source:
URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC11244353/
Title: Shot Refinement via Fusion of Shuttlecock Tracking and Hit Detection from a Monocular Camera (PMC11244353)
Found via: classical/lightweight search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
   - be a concrete, checkable statement (not vague generalities)
   - include a direct quote from the source as support
   - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-06T17:30:30.490Z)

I'll fetch the source and extract the relevant claims.

3. user (2026-06-06T17:30:45.111Z)

Badminton Shot Detection via Shuttlecock Tracking and Hit Detection

Hit/Shot Detection Method

The paper proposes a **Shot Refinement Algorithm (SRA)** that combines two detection approaches:

1. **Hit Detection by Trajectories (HD-T)**: Uses TrackNet to identify sudden direction changes in the shuttlecock
2. **Hit Detection by Action (HD-A)**: Uses YOLOv7 to detect player swing actions

The SRA merges these outputs by creating "Hit Segments" from action detections, then validates trajectory detections against player actions. As stated: "the intuition of SRA is to combine the outputs of both modules and produce fine-grained results."

Tracking Models Used

- **TrackNet**: A deep neural network for tracking small, fast-moving objects
- **YOLOv7**: Used for detecting player swinging actions
- **DensePose**: For player pose estimation

Reported Performance Metrics

For shot extraction (IoU threshold 0.5):
- **Precision**: 89.7% (tuned HD-T + SRA)
- **Recall**: 91.3%
- **F1 Score**: 90.5%

TrackNet alone achieved only 58.8% precision, 93.6% recall, and 72.3% F1.

For shot-type classification: 72.1% accuracy at IoU threshold 0.95.

Camera Setup

"Single-player badminton matches" analyzed from "a monocular camera." The system requires only standard video without additional hardware.

Datasets

- **BWF dataset**: 32 matches, 602 rallies, 8,975 images from Badminton World Federation YouTube videos
- **NUCIA dataset**: 354 matches, 6,855 rallies, 98,675 images from Taiwan Ministry of Education competition

Compute Requirements

"10th generation Core i9 Intel CPU with 32 GB of RAM and an RTX 3080Ti GPU." Processing time: 0.174 seconds per frame with SRA.

Publication Details

Published July 5, 2024 in *Sensors* (Basel, Switzerland), volume 24, issue 13, article 4372. Authors: Yi-Hua Hsu, Chih-Chang Yu, Hsu-Yung Cheng.

4. assistant (2026-06-06T17:30:48.514Z)

I have the key details. Let me note the licensing question for *Sensors* (MDPI) which is open access (CC-BY), and verify a couple of specifics. The fetch was successful and the content is highly relevant. Let me compile the structured output.

5. user (2026-06-06T17:30:57.701Z)

Structured output provided successfully

6. assistant (2026-06-06T17:31:08.435Z)

I've extracted and returned the key claims from the source.

Summary of findings: This is a primary peer-reviewed research paper (Hsu, Yu, Cheng; *Sensors* 24(13):4372, MDPI, published 2024-07-05). MDPI *Sensors* is open-access under CC-BY, so the paper itself is freely reusable, though the BWF/NUCIA datasets it uses are YouTube/competition-sourced and not clearly permissively licensed.

Most relevant to the research question: the paper directly validates a **multi-modal fusion** strategy that maps cleanly onto our available local inputs (shuttle trajectory + person/pose detector). Their Shot Refinement Algorithm fuses TrackNet trajectory-based hit detection (HD-T) with YOLOv7 player swing-action detection (HD-A), lifting shot-extraction precision from 58.8% (trajectory-only) to 89.7% and F1 from 72.3% to 90.5% at IoU 0.5. The key takeaway for our setup: **trajectory-only hit detection is recall-strong but precision-weak (58.8%), and adding a player-action modality is what closes the precision gap** ? a strong, directly-actionable signal that our WASB-trajectory pipeline could be complemented by player pose/action detection rather than (or alongside) the Gemini-teacher distillation path. Runs on a single consumer GPU (RTX 3080Ti) at 0.174 s/frame from a monocular camera, consistent with our local-first single-GPU constraint.

Note: YOLOv7 carries a GPL-3.0 license, which is a commercial-viability caveat we'd want to flag if adopting their exact action-detection component ? though the fusion *concept* could be reimplemented with our permissive torchvision/pose substrate.